Assignment 8

Border Gateway Protocol (BGP)

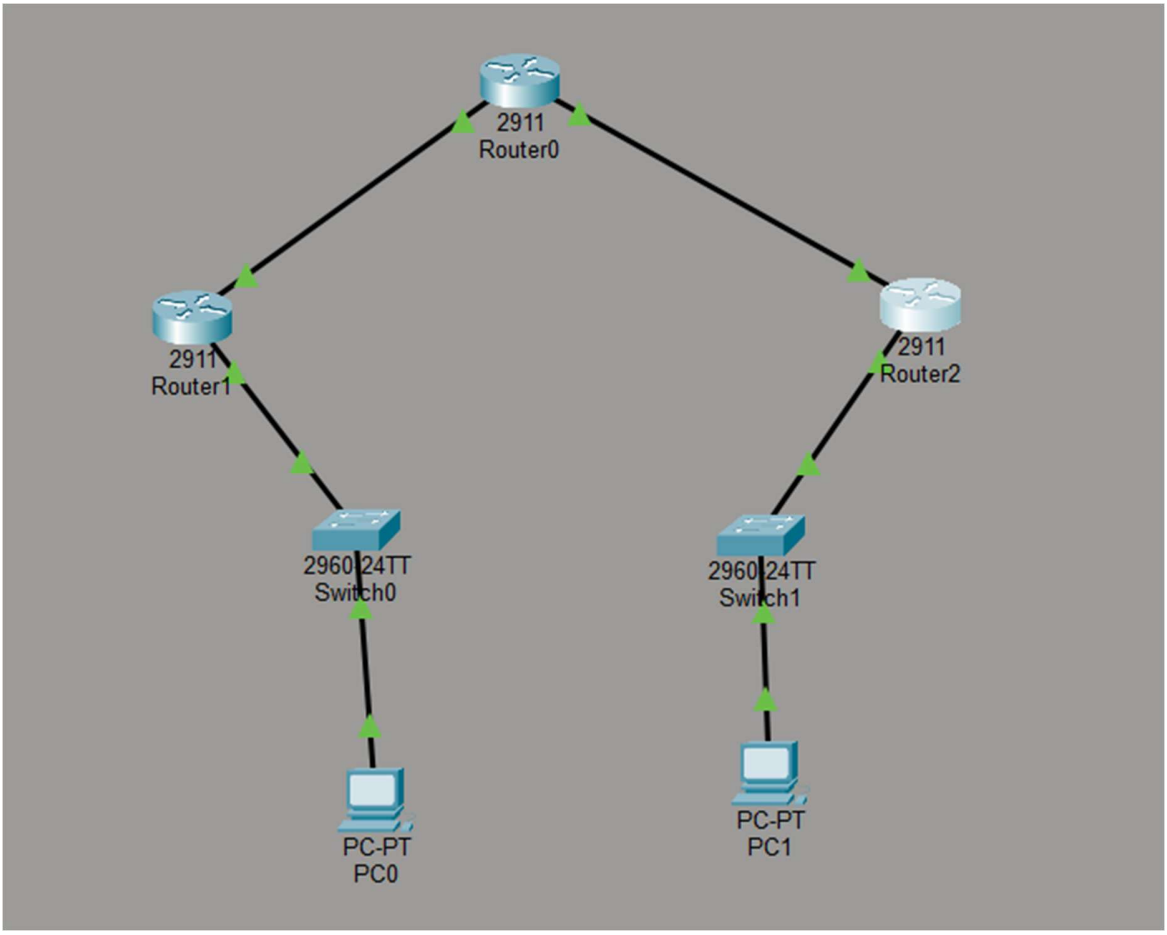
Configuration and Analysis of BGP Routing

Objective

- To understand the working of Border Gateway Protocol (BGP)
- To configure External BGP (eBGP) between different Autonomous Systems
- To analyze path selection using BGP attributes

Output

1: Topology — All Links Green



2: show ip bgp summary — Router0 (R2)

```
Router#show ip bgp summary
BGP router identifier 10.0.0.5, local AS number 200
BGP table version is 3, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 632 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.1        4   100       0       0        3    0    0 00:07:17        4
10.0.0.6        4   300       6       5        3    0    0 00:03:16        4
```

3: show ip bgp — BGP Routing Table

```
Router#
Router#show ip bgp
BGP table version is 4, local router ID is 10.0.0.5
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.0.0/30       0.0.0.0                  0     0 32768 i
*> 192.168.3.0/24    10.0.0.6                 0     0     0 300 i
```

4: show ip route — B Routes

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
C       10.0.0.4/30 is directly connected, GigabitEthernet0/1
L       10.0.0.5/32 is directly connected, GigabitEthernet0/1
B    192.168.3.0/24 [20/0] via 10.0.0.6, 00:00:00
```

5: Ping PC0 → PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time<1ms TTL=128
Reply from 192.168.30.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```